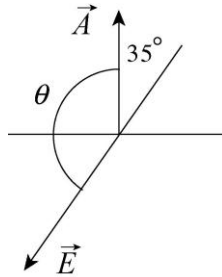


Chapter 23

1. **THINK** This exercise deals with electric flux through a square surface.

EXPRESS The vector area $\vec{A}$ and the electric field $\vec{E}$ are shown on the diagram below.



The electric flux through the surface is given by $\Phi = \vec{E} \cdot \vec{A} = EA \cos \theta$.

EXPRESS The angle θ between $\vec{A}$ and $\vec{E}$ is $180^\circ - 35^\circ = 145^\circ$, so the electric flux through the area is

$$\Phi = EA \cos \theta = (1800 \text{ N/C})(3.2 \times 10^{-3} \text{ m})^2 \cos 145^\circ = -1.5 \times 10^{-2} \text{ N} \cdot \text{m}^2/\text{C}.$$

LEARN The flux is a maximum when $\vec{A}$ and $\vec{E}$ points in the same direction ($\theta = 0$), and is zero when the two vectors are perpendicular to each other ($\theta = 90$).

2. We use $\Phi = \oint \vec{E} \cdot d\vec{A}$ and note that the side length of the cube is $(3.0 \text{ m} - 1.0 \text{ m}) = 2.0 \text{ m}$.

(a) On the top face of the cube $y = 2.0 \text{ m}$ and $d\vec{A} = (dA)\hat{j}$. Therefore, we have

$$\vec{E} = 4\hat{i} - 3((2.0)^2 + 2)\hat{j} = 4\hat{i} - 18\hat{j}. \text{ Thus the flux is}$$

$$\Phi = \int_{\text{top}} \vec{E} \cdot d\vec{A} = \int_{\text{top}} (4\hat{i} - 18\hat{j}) \cdot (dA)\hat{j} = -18 \int_{\text{top}} dA = (-18)(2.0)^2 \text{ N} \cdot \text{m}^2/\text{C} = -72 \text{ N} \cdot \text{m}^2/\text{C}.$$

(b) On the bottom face of the cube $y = 0$ and $d\vec{A} = (dA)(-\hat{j})$. Therefore, we have

$$\vec{E} = 4\hat{i} - 3(0^2 + 2)\hat{j} = 4\hat{i} - 6\hat{j}. \text{ Thus, the flux is}$$

$$\Phi = \int_{\text{bottom}} \vec{E} \cdot d\vec{A} = \int_{\text{bottom}} (4\hat{i} - 6\hat{j}) \cdot (dA)(-\hat{j}) = 6 \int_{\text{bottom}} dA = 6(2.0)^2 \text{ N} \cdot \text{m}^2/\text{C} = +24 \text{ N} \cdot \text{m}^2/\text{C}.$$

(c) On the left face of the cube $d\vec{A} = (dA)(-\hat{i})$. So

$$\Phi = \int_{\text{left}} \vec{E} \cdot d\vec{A} = \int_{\text{left}} (4\hat{i} + E_y\hat{j}) \cdot (dA)(-\hat{i}) = -4 \int_{\text{bottom}} dA = -4(2.0)^2 \text{ N} \cdot \text{m}^2/\text{C} = -16 \text{ N} \cdot \text{m}^2/\text{C}.$$

(d) On the back face of the cube $d\vec{A} = (dA)(-\hat{k})$. But since $\vec{E}$ has no z component $\vec{E} \cdot d\vec{A} = 0$. Thus, $\Phi = 0$.

(e) We now have to add the flux through all six faces. One can easily verify that the flux through the front face is zero, while that through the right face is the opposite of that through the left one, or $+16 \text{ N} \cdot \text{m}^2/\text{C}$. Thus the net flux through the cube is

$$\Phi = (-72 + 24 - 16 + 0 + 0 + 16) \text{ N} \cdot \text{m}^2/\text{C} = -48 \text{ N} \cdot \text{m}^2/\text{C}.$$

3. We use $\Phi = \vec{E} \cdot \vec{A}$, where $\vec{A} = A\hat{j} = (1.40 \text{ m})^2 \hat{j}$.

$$(a) \Phi = (6.00 \text{ N/C})\hat{i} \cdot (1.40 \text{ m})^2 \hat{j} = 0.$$

$$(b) \Phi = (-2.00 \text{ N/C})\hat{j} \cdot (1.40 \text{ m})^2 \hat{j} = -3.92 \text{ N} \cdot \text{m}^2/\text{C}.$$

$$(c) \Phi = [(-3.00 \text{ N/C})\hat{i} + (400 \text{ N/C})\hat{k}] \cdot (1.40 \text{ m})^2 \hat{j} = 0.$$

(d) The total flux of a uniform field through a closed surface is always zero.

4. The flux through the flat surface encircled by the rim is given by $\Phi = \pi a^2 E$. Thus, the flux through the netting is

$$\Phi' = -\Phi = -\pi a^2 E = -\pi(0.11 \text{ m})^2(3.0 \times 10^{-3} \text{ N/C}) = -1.1 \times 10^{-4} \text{ N} \cdot \text{m}^2/\text{C}.$$

5. To exploit the symmetry of the situation, we imagine a closed Gaussian surface in the shape of a cube, of edge length d , with a proton of charge $q = +1.6 \times 10^{-19} \text{ C}$ situated at the inside center of the cube. The cube has six faces, and we expect an equal amount of flux through each face. The total amount of flux is $\Phi_{\text{net}} = q/\epsilon_0$, and we conclude that the flux through the square is one-sixth of that. Thus,

$$\Phi = \frac{q}{6\epsilon_0} = \frac{1.6 \times 10^{-19} \text{ C}}{6(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 3.01 \times 10^{-9} \text{ N} \cdot \text{m}^2/\text{C}.$$

6. There is no flux through the sides, so we have two “inward” contributions to the flux, one from the top (of magnitude $(34)(3.0)^2$) and one from the bottom (of magnitude

(20)(3.0)²). With “inward” flux being negative, the result is $\Phi = -486 \text{ N}\cdot\text{m}^2/\text{C}$. Gauss’ law then leads to

$$q_{\text{enc}} = \epsilon_0 \Phi = (8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2)(-486 \text{ N}\cdot\text{m}^2/\text{C}) = -4.3 \times 10^{-9} \text{ C}.$$

7. We use Gauss’ law: $\epsilon_0 \Phi = q$, where Φ is the total flux through the cube surface and q is the net charge inside the cube. Thus,

$$\Phi = \frac{q}{\epsilon_0} = \frac{1.8 \times 10^{-6} \text{ C}}{8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2} = 2.0 \times 10^5 \text{ N}\cdot\text{m}^2/\text{C}.$$

8. (a) The total surface area bounding the bathroom is

$$A = 2(2.5 \times 3.0) + 2(3.0 \times 2.0) + 2(2.0 \times 2.5) = 37 \text{ m}^2.$$

The absolute value of the total electric flux, with the assumptions stated in the problem, is

$$|\Phi| = \left| \sum \vec{E} \cdot \vec{A} \right| = |\vec{E}| A = (600 \text{ N/C})(37 \text{ m}^2) = 22 \times 10^3 \text{ N}\cdot\text{m}^2/\text{C}.$$

By Gauss’ law, we conclude that the enclosed charge (in absolute value) is $|q_{\text{enc}}| = \epsilon_0 |\Phi| = 2.0 \times 10^{-7} \text{ C}$. Therefore, with volume $V = 15 \text{ m}^3$, and recognizing that we are dealing with negative charges, the charge density is

$$\rho = \frac{q_{\text{enc}}}{V} = \frac{-2.0 \times 10^{-7} \text{ C}}{15 \text{ m}^3} = -1.3 \times 10^{-8} \text{ C/m}^3.$$

(b) We find $(|q_{\text{enc}}|/e)/V = (2.0 \times 10^{-7} \text{ C}/1.6 \times 10^{-19} \text{ C})/15 \text{ m}^3 = 8.2 \times 10^{10}$ excess electrons per cubic meter.

9. (a) Let $A = (1.40 \text{ m})^2$. Then

$$\Phi = (3.00y \hat{j}) \cdot (-A \hat{j}) \Big|_{y=0} + (3.00y \hat{j}) \cdot (A \hat{j}) \Big|_{y=1.40} = (3.00)(1.40)(1.40)^2 = 8.23 \text{ N}\cdot\text{m}^2/\text{C}.$$

(b) The charge is given by

$$q_{\text{enc}} = \epsilon_0 \Phi = (8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2)(8.23 \text{ N}\cdot\text{m}^2/\text{C}) = 7.29 \times 10^{-11} \text{ C}.$$

(c) The electric field can be re-written as $\vec{E} = 3.00y \hat{j} + \vec{E}_0$, where $\vec{E}_0 = -4.00\hat{i} + 6.00\hat{j}$ is a constant field which does not contribute to the net flux through the cube. Thus Φ is still $8.23 \text{ N}\cdot\text{m}^2/\text{C}$.

(d) The charge is again given by

$$q_{\text{enc}} = \epsilon_0 \Phi = (8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2) (8.23 \text{ N} \cdot \text{m}^2 / \text{C}) = 7.29 \times 10^{-11} \text{ C}.$$

10. None of the constant terms will result in a nonzero contribution to the flux (see Eq. 23-4 and Eq. 23-7), so we focus on the x dependent term only. In SI units, we have

$$E_{\text{nonconstant}} = 3x \hat{i}.$$

The face of the cube located at $x = 0$ (in the yz plane) has area $A = 4 \text{ m}^2$ (and it “faces” the $+\hat{i}$ direction) and has a “contribution” to the flux equal to $E_{\text{nonconstant}} A = (3)(0)(4) = 0$. The face of the cube located at $x = -2 \text{ m}$ has the same area A (and this one “faces” the $-\hat{i}$ direction) and a contribution to the flux:

$$-E_{\text{nonconstant}} A = -(3)(-2)(4) = 24 \text{ N} \cdot \text{m} / \text{C}^2.$$

Thus, the net flux is $\Phi = 0 + 24 = 24 \text{ N} \cdot \text{m} / \text{C}^2$. According to Gauss’ law, we therefore have $q_{\text{enc}} = \epsilon_0 \Phi = 2.13 \times 10^{-10} \text{ C}$.

11. None of the constant terms will result in a nonzero contribution to the flux (see Eq. 23-4 and Eq. 23-7), so we focus on the x dependent term only:

$$E_{\text{nonconstant}} = (-4.00y^2) \hat{i} \text{ (in SI units)}.$$

The face of the cube located at $y = 4.00$ has area $A = 4.00 \text{ m}^2$ (and it “faces” the $+\hat{j}$ direction) and has a “contribution” to the flux equal to

$$E_{\text{nonconstant}} A = (-4)(4^2)(4) = -256 \text{ N} \cdot \text{m} / \text{C}^2.$$

The face of the cube located at $y = 2.00 \text{ m}$ has the same area A (however, this one “faces” the $-\hat{j}$ direction) and a contribution to the flux:

$$-E_{\text{nonconstant}} A = -(-4)(2^2)(4) = 64 \text{ N} \cdot \text{m} / \text{C}^2.$$

Thus, the net flux is $\Phi = (-256 + 64) \text{ N} \cdot \text{m} / \text{C}^2 = -192 \text{ N} \cdot \text{m} / \text{C}^2$. According to Gauss’s law, we therefore have

$$q_{\text{enc}} = \epsilon_0 \Phi = (8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2) (-192 \text{ N} \cdot \text{m}^2 / \text{C}) = -1.70 \times 10^{-9} \text{ C}.$$

12. We note that only the smaller shell contributes a (nonzero) field at the designated point, since the point is inside the radius of the large sphere (and $E = 0$ inside of a spherical charge), and the field points toward the $-x$ direction. Thus, with $R = 0.020 \text{ m}$ (the radius of the smaller shell), $L = 0.10 \text{ m}$ and $x = 0.020 \text{ m}$, we obtain

$$\begin{aligned}\vec{E} &= E(-\hat{j}) = -\frac{q}{4\pi\epsilon_0 r^2} \hat{j} = -\frac{4\pi R^2 \sigma_2}{4\pi\epsilon_0 (L-x)^2} \hat{j} = -\frac{R^2 \sigma_2}{\epsilon_0 (L-x)^2} \hat{j} \\ &= -\frac{(0.020 \text{ m})^2 (4.0 \times 10^{-6} \text{ C/m}^2)}{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(0.10 \text{ m} - 0.020 \text{ m})^2} \hat{j} = (-2.8 \times 10^4 \text{ N/C}) \hat{j}.\end{aligned}$$

13. **THINK** A cube has six surfaces. The total flux through the cube is the sum of fluxes through each individual surface. We use Gauss' law to find the net charge inside the cube.

EXPRESS Let A be the area of one face of the cube, E_u be the magnitude of the electric field at the upper face, and E_l be the magnitude of the field at the lower face. Since the field is downward, the flux through the upper face is negative and the flux through the lower face is positive. The flux through the other faces is zero (because their area vectors are parallel to the field), so the total flux through the cube surface is

$$\Phi = A(E_l - E_u).$$

The net charge inside the cube is given by Gauss' law: $q = \epsilon_0 \Phi$.

ANALYZE Substituting the values given, we find the net charge to be

$$\begin{aligned}q &= \epsilon_0 \Phi = \epsilon_0 A(E_l - E_u) = (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(100 \text{ m})^2(100 \text{ N/C} - 60.0 \text{ N/C}) \\ &= 3.54 \times 10^{-6} \text{ C} = 3.54 \mu\text{C}.\end{aligned}$$

LEARN Since $\Phi > 0$, we conclude that the cube encloses a net positive charge.

14. Equation 23-6 (Gauss' law) gives $\epsilon_0 \Phi = q_{\text{enc}}$.

(a) Thus, the value $\Phi = 2.0 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$ for small r leads to

$$q_{\text{central}} = \epsilon_0 \Phi = (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(2.0 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}) = 1.77 \times 10^{-6} \text{ C} \approx 1.8 \times 10^{-6} \text{ C}.$$

(b) The next value that Φ takes is $\Phi = -4.0 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$, which implies that $q_{\text{enc}} = -3.54 \times 10^{-6} \text{ C}$. But we have already accounted for some of that charge in part (a), so the result for part (b) is

$$q_A = q_{\text{enc}} - q_{\text{central}} = -5.3 \times 10^{-6} \text{ C}.$$

(c) Finally, the large r value for Φ is $\Phi = 6.0 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$, which implies that $q_{\text{total enc}} = 5.31 \times 10^{-6} \text{ C}$. Considering what we have already found, then the result is $q_{\text{total enc}} - q_A - q_{\text{central}} = +8.9 \mu\text{C}$.

15. The total flux through any surface that completely surrounds the point charge is q/ϵ_0 .

(a) If we stack identical cubes side by side and directly on top of each other, we will find that eight cubes meet at any corner. Thus, one-eighth of the field lines emanating from the point charge pass through a cube with a corner at the charge, and the total flux through the surface of such a cube is $q/8\epsilon_0$. Now the field lines are radial, so at each of the three cube faces that meet at the charge, the lines are parallel to the face and the flux through the face is zero.

(b) The fluxes through each of the other three faces are the same, so the flux through each of them is one-third of the total. That is, the flux through each of these faces is $(1/3)(q/8\epsilon_0) = q/24\epsilon_0$. Thus, the multiple is $1/24 = 0.0417$.

16. The total electric flux through the cube is $\Phi = \oint \vec{E} \cdot d\vec{A}$. The net flux through the two faces parallel to the yz plane is

$$\begin{aligned}\Phi_{yz} &= \iint [E_x(x=x_2) - E_x(x=x_1)] dydz = \int_{y_1=0}^{y_2=1} dy \int_{z_1=1}^{z_2=3} dz [10 + 2(4) - 10 - 2(1)] \\ &= 6 \int_{y_1=0}^{y_2=1} dy \int_{z_1=1}^{z_2=3} dz = 6(1)(2) = 12.\end{aligned}$$

Similarly, the net flux through the two faces parallel to the xz plane is

$$\Phi_{xz} = \iint [E_y(y=y_2) - E_y(y=y_1)] dx dz = \int_{x_1=1}^{x_2=4} dx \int_{z_1=1}^{z_2=3} dz [-3 - (-3)] = 0,$$

and the net flux through the two faces parallel to the xy plane is

$$\Phi_{xy} = \iint [E_z(z=z_2) - E_z(z=z_1)] dx dy = \int_{x_1=1}^{x_2=4} dx \int_{y_1=0}^{y_2=1} dy (3b - b) = 2b(3)(1) = 6b.$$

Applying Gauss' law, we obtain

$$q_{\text{enc}} = \epsilon_0 \Phi = \epsilon_0 (\Phi_{xy} + \Phi_{xz} + \Phi_{yz}) = \epsilon_0 (6.00b + 0 + 12.0) = 24.0\epsilon_0$$

which implies that $b = 2.00 \text{ N/C} \cdot \text{m}$.

17. **THINK** The system has spherical symmetry, so our Gaussian surface is a sphere of radius R with a surface area $A = 4\pi R^2$.

EXPRESS The charge on the surface of the sphere is the product of the surface charge density σ and the surface area of the sphere: $q = \sigma A = \sigma(4\pi R^2)$. We calculate the total electric flux leaving the surface of the sphere using Gauss' law: $q = \epsilon_0 \Phi$.

ANALYZE (a) With $R = (1.20 \text{ m})/2 = 0.60 \text{ m}$ and $\sigma = 8.1 \times 10^{-6} \text{ C/m}^2$, the charge on the surface is

$$q = 4\pi R^2 \sigma = 4\pi (0.60 \text{ m})^2 (8.1 \times 10^{-6} \text{ C/m}^2) = 3.7 \times 10^{-5} \text{ C}.$$

(b) We choose a Gaussian surface in the form of a sphere, concentric with the conducting sphere and with a slightly larger radius. By Gauss's law, the flux is

$$\Phi = \frac{q}{\epsilon_0} = \frac{3.66 \times 10^{-5} \text{ C}}{8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2} = 4.1 \times 10^6 \text{ N} \cdot \text{m}^2 / \text{C}.$$

LEARN Since there is no charge inside the conducting sphere, the total electric flux through the surface of the sphere only depends on the charge residing on the surface of the sphere.

18. Using Eq. 23-11, the surface charge density is

$$\sigma = E\epsilon_0 = (2.3 \times 10^5 \text{ N/C})(8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2) = 2.0 \times 10^{-6} \text{ C/m}^2.$$

19. (a) The area of a sphere may be written $4\pi R^2 = \pi D^2$. Thus,

$$\sigma = \frac{q}{\pi D^2} = \frac{2.4 \times 10^{-6} \text{ C}}{\pi (1.3 \text{ m})^2} = 4.5 \times 10^{-7} \text{ C/m}^2.$$

(b) Equation 23-11 gives

$$E = \frac{\sigma}{\epsilon_0} = \frac{4.5 \times 10^{-7} \text{ C/m}^2}{8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2} = 5.1 \times 10^4 \text{ N/C}.$$

20. Equation 23-6 (Gauss' law) gives $\epsilon_0 \Phi = q_{\text{enc}}$.

(a) The value $\Phi = -9.0 \times 10^5 \text{ N} \cdot \text{m}^2 / \text{C}$ for small r leads to $q_{\text{central}} = -7.97 \times 10^{-6} \text{ C}$ or roughly $-8.0 \mu\text{C}$.

(b) The next (nonzero) value that Φ takes is $\Phi = +4.0 \times 10^5 \text{ N} \cdot \text{m}^2 / \text{C}$, which implies $q_{\text{enc}} = 3.54 \times 10^{-6} \text{ C}$. But we have already accounted for some of that charge in part (a), so the result is

$$q_A = q_{\text{enc}} - q_{\text{central}} = 11.5 \times 10^{-6} \text{ C} \approx 12 \mu\text{C}.$$

(c) Finally, the large r value for Φ is $\Phi = -2.0 \times 10^5 \text{ N} \cdot \text{m}^2 / \text{C}$, which implies $q_{\text{total enc}} = -1.77 \times 10^{-6} \text{ C}$. Considering what we have already found, then the result is

$$q_{\text{total enc}} - q_A - q_{\text{central}} = -5.3 \mu\text{C}.$$

21. (a) Consider a Gaussian surface that is completely within the conductor and surrounds the cavity. Since the electric field is zero everywhere on the surface, the net charge it

encloses is zero. The net charge is the sum of the charge q in the cavity and the charge q_w on the cavity wall, so $q + q_w = 0$ and $q_w = -q = -3.0 \times 10^{-6} \text{ C}$.

(b) The net charge Q of the conductor is the sum of the charge on the cavity wall and the charge q_s on the outer surface of the conductor, so $Q = q_w + q_s$ and

$$q_s = Q - q_w = (10 \times 10^{-6} \text{ C}) - (-3.0 \times 10^{-6} \text{ C}) = +1.3 \times 10^{-5} \text{ C}.$$

22. We combine Newton's second law ($F = ma$) with the definition of electric field ($F = qE$) and with Eq. 23-12 (for the field due to a line of charge). In terms of magnitudes, we have (if $r = 0.080 \text{ m}$ and $\lambda = 6.0 \times 10^{-6} \text{ C/m}$)

$$ma = eE = \frac{e\lambda}{2\pi\epsilon_0 r} \Rightarrow a = \frac{e\lambda}{2\pi\epsilon_0 r m} = 2.1 \times 10^{17} \text{ m/s}^2.$$

23. (a) The side surface area A for the drum of diameter D and length h is given by $A = \pi Dh$. Thus,

$$\begin{aligned} q &= \sigma A = \sigma \pi Dh = \pi \epsilon_0 EDh \\ &= \pi (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2) (2.3 \times 10^5 \text{ N/C}) (0.12 \text{ m}) (0.42 \text{ m}) \\ &= 3.2 \times 10^{-7} \text{ C}. \end{aligned}$$

(b) The new charge is

$$q' = q \left(\frac{A'}{A} \right) = q \left(\frac{\pi D' h'}{\pi Dh} \right) = (3.2 \times 10^{-7} \text{ C}) \left[\frac{(8.0 \text{ cm})(28 \text{ cm})}{(12 \text{ cm})(42 \text{ cm})} \right] = 1.4 \times 10^{-7} \text{ C}.$$

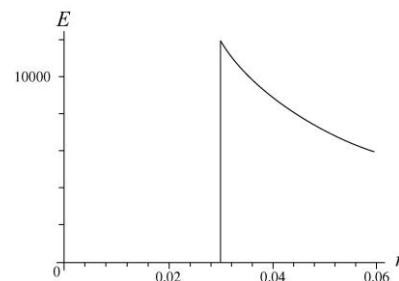
24. We imagine a cylindrical Gaussian surface A of radius r and unit length concentric with the metal tube. Then by symmetry $\oint_A \vec{E} \cdot d\vec{A} = 2\pi r E = \frac{q_{\text{enc}}}{\epsilon_0}$.

(a) For $r < R$, $q_{\text{enc}} = 0$, so $E = 0$.

(b) For $r > R$, $q_{\text{enc}} = \lambda$, so $E(r) = \lambda / 2\pi r \epsilon_0$. With $\lambda = 2.00 \times 10^{-8} \text{ C/m}$ and $r = 2.00R = 0.0600 \text{ m}$, we obtain

$$E = \frac{(2.0 \times 10^{-8} \text{ C/m})}{2\pi (0.0600 \text{ m}) (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 5.99 \times 10^3 \text{ N/C}.$$

(c) The plot of E vs. r is shown to the right. Here, the maximum value is



$$E_{\max} = \frac{\lambda}{2\pi r \epsilon_0} = \frac{(2.0 \times 10^{-8} \text{ C/m})}{2\pi(0.030 \text{ m})(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 1.2 \times 10^4 \text{ N/C}.$$

25. **THINK** Our system is an infinitely long line of charge. Since the system possesses cylindrical symmetry, we may apply Gauss' law and take the Gaussian surface to be in the form of a closed cylinder.

EXPRESS We imagine a cylindrical Gaussian surface A of radius r and length h concentric with the metal tube. Then by symmetry,

$$\oint_A \vec{E} \cdot d\vec{A} = 2\pi r h E = \frac{q}{\epsilon_0},$$

where q is the amount of charge enclosed by the Gaussian cylinder. Thus, the magnitude of the electric field produced by a uniformly charged infinite line is

$$E = \frac{q/h}{2\pi\epsilon_0 r} = \frac{\lambda}{2\pi\epsilon_0 r}$$

where λ is the linear charge density and r is the distance from the line to the point where the field is measured.

ANALYZE Substituting the values given, we have

$$\begin{aligned} \lambda &= 2\pi\epsilon_0 E r = 2\pi(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(4.5 \times 10^4 \text{ N/C})(2.0 \text{ m}) \\ &= 5.0 \times 10^{-6} \text{ C/m}. \end{aligned}$$

LEARN Since $\lambda > 0$, the direction of $\vec{E}$ is radially outward from the line of charge. Note that the field varies with r as $E \sim 1/r$, in contrast to the $1/r^2$ dependence due to a point charge.

26. As we approach $r = 3.5 \text{ cm}$ from the inside, we have

$$E_{\text{internal}} = \frac{2\lambda}{4\pi\epsilon_0 r} = 1000 \text{ N/C}.$$

And as we approach $r = 3.5 \text{ cm}$ from the outside, we have

$$E_{\text{external}} = \frac{2\lambda}{4\pi\epsilon_0 r} + \frac{2\lambda'}{4\pi\epsilon_0 r} = -3000 \text{ N/C}.$$

Considering the difference ($E_{\text{external}} - E_{\text{internal}}$) allows us to find λ' (the charge per unit length on the larger cylinder). Using $r = 0.035$ m, we obtain $\lambda' = -5.8 \times 10^{-9}$ C/m.

27. We denote the radius of the thin cylinder as $R = 0.015$ m. Using Eq. 23-12, the net electric field for $r > R$ is given by

$$E_{\text{net}} = E_{\text{wire}} + E_{\text{cylinder}} = \frac{-\lambda}{2\pi\epsilon_0 r} + \frac{\lambda'}{2\pi\epsilon_0 r}$$

where $-\lambda = -3.6$ nC/m is the linear charge density of the wire and λ' is the linear charge density of the thin cylinder. We note that the surface and linear charge densities of the thin cylinder are related by

$$q_{\text{cylinder}} = \lambda' L = \sigma(2\pi RL) \Rightarrow \lambda' = \sigma(2\pi R).$$

Now, E_{net} outside the cylinder will equal zero, provided that $2\pi R\sigma = \lambda$, or

$$\sigma = \frac{\lambda}{2\pi R} = \frac{3.6 \times 10^{-6} \text{ C/m}}{(2\pi)(0.015 \text{ m})} = 3.8 \times 10^{-8} \text{ C/m}^2.$$

28. (a) In Eq. 23-12, $\lambda = q/L$ where q is the net charge enclosed by a cylindrical Gaussian surface of radius r . The field is being measured outside the system (the charged rod coaxial with the neutral cylinder) so that the net enclosed charge is only that which is on the rod. Consequently,

$$|\vec{E}| = \frac{2\lambda}{4\pi\epsilon_0 r} = \frac{2(2.0 \times 10^{-9} \text{ C/m})}{4\pi\epsilon_0 (0.15 \text{ m})} = 2.4 \times 10^2 \text{ N/C}.$$

(b) Since the field is zero inside the conductor (in an electrostatic configuration), then there resides on the inner surface charge $-q$, and on the outer surface, charge $+q$ (where q is the charge on the rod at the center). Therefore, with $r_i = 0.05$ m, the surface density of charge is

$$\sigma_{\text{inner}} = \frac{-q}{2\pi r_i L} = -\frac{\lambda}{2\pi r_i} = -\frac{2.0 \times 10^{-9} \text{ C/m}}{2\pi(0.050 \text{ m})} = -6.4 \times 10^{-9} \text{ C/m}^2$$

for the inner surface.

(c) With $r_o = 0.10$ m, the surface charge density of the outer surface is

$$\sigma_{\text{outer}} = \frac{+q}{2\pi r_o L} = \frac{\lambda}{2\pi r_o} = +3.2 \times 10^{-9} \text{ C/m}^2.$$

29. **THINK** The charge densities of both the conducting cylinder and the shell are uniform, and we neglect fringing effect. Symmetry can be used to show that the electric field is radial, both between the cylinder and the shell and outside the shell. It is zero, of course, inside the cylinder and inside the shell.

EXPRESS We take the Gaussian surface to be a cylinder of length L , coaxial with the given cylinders and of radius r . The flux through this surface is $\Phi = 2\pi rLE$, where E is the magnitude of the field at the Gaussian surface. We may ignore any flux through the ends. Gauss' law yields $q_{\text{enc}} = \epsilon_0 \Phi = 2\pi r\epsilon_0 LE$, where q_{enc} is the charge enclosed by the Gaussian surface.

ANALYZE (a) In this case, we take the radius of our Gaussian cylinder to be

$$r = 2.00R_2 = 20.0R_1 = (20.0)(1.3 \times 10^{-3} \text{ m}) = 2.6 \times 10^{-2} \text{ m}.$$

The charge enclosed is

$$q_{\text{enc}} = Q_1 + Q_2 = -Q_1 = -3.40 \times 10^{-12} \text{ C}.$$

Consequently, Gauss' law yields

$$E = \frac{q_{\text{enc}}}{2\pi\epsilon_0 Lr} = \frac{-3.40 \times 10^{-12} \text{ C}}{2\pi(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(11.0 \text{ m})(2.60 \times 10^{-2} \text{ m})} = -0.214 \text{ N/C},$$

or $|E| = 0.214 \text{ N/C}$.

(b) The negative sign in E indicates that the field points inward.

(c) Next, for $r = 5.00 R_1$, the charge enclosed by the Gaussian surface is $q_{\text{enc}} = Q_1 = 3.40 \times 10^{-12} \text{ C}$. Consequently, Gauss' law yields $2\pi r\epsilon_0 LE = q_{\text{enc}}$, or

$$E = \frac{q_{\text{enc}}}{2\pi\epsilon_0 Lr} = \frac{3.40 \times 10^{-12} \text{ C}}{2\pi(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(11.0 \text{ m})(5.00 \times 1.30 \times 10^{-3} \text{ m})} = 0.855 \text{ N/C}.$$

(d) The positive sign indicates that the field points outward.

(e) We consider a cylindrical Gaussian surface whose radius places it within the shell itself. The electric field is zero at all points on the surface since any field within a conducting material would lead to current flow (and thus to a situation other than the electrostatic ones being considered here), so the total electric flux through the Gaussian surface is zero and the net charge within it is zero (by Gauss' law). Since the central rod has charge Q_1 , the inner surface of the shell must have charge $Q_{\text{in}} = -Q_1 = -3.40 \times 10^{-12} \text{ C}$.

(f) Since the shell is known to have total charge $Q_2 = -2.00Q_1$, it must have charge $Q_{\text{out}} = Q_2 - Q_{\text{in}} = -Q_1 = -3.40 \times 10^{-12} \text{ C}$ on its outer surface.

LEARN Cylindrical symmetry of the system allows us to apply Gauss' law to the problem. Since electric field is zero inside the conducting shell, by Gauss' law, any net charge must be distributed on the surfaces of the shells.

30. We reason that point P (the point on the x axis where the net electric field is zero) cannot be between the lines of charge (since their charges have opposite sign). We reason further that P is not to the left of "line 1" since its magnitude of charge (per unit length) exceeds that of "line 2"; thus, we look in the region to the right of "line 2" for P . Using Eq. 23-12, we have

$$E_{\text{net}} = E_1 + E_2 = \frac{2\lambda_1}{4\pi\epsilon_0(x+L/2)} + \frac{2\lambda_2}{4\pi\epsilon_0(x-L/2)}.$$

Setting this equal to zero and solving for x we find

$$x = \left(\frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2} \right) \frac{L}{2} = \left(\frac{6.0\mu\text{C/m} - (-2.0\mu\text{C/m})}{6.0\mu\text{C/m} + (-2.0\mu\text{C/m})} \right) \frac{8.0\text{ cm}}{2} = 8.0\text{ cm}.$$

31. We denote the inner and outer cylinders with subscripts i and o , respectively.

(a) Since $r_i < r = 4.0\text{ cm} < r_o$,

$$E(r) = \frac{\lambda_i}{2\pi\epsilon_0 r} = \frac{5.0 \times 10^{-6}\text{ C/m}}{2\pi(8.85 \times 10^{-12}\text{ C}^2/\text{N}\cdot\text{m}^2)(4.0 \times 10^{-2}\text{ m})} = 2.3 \times 10^6\text{ N/C}.$$

(b) The electric field $\vec{E}(r)$ points radially outward.

(c) Since $r > r_o$,

$$E(r = 8.0\text{ cm}) = \frac{\lambda_i + \lambda_o}{2\pi\epsilon_0 r} = \frac{5.0 \times 10^{-6}\text{ C/m} - 7.0 \times 10^{-6}\text{ C/m}}{2\pi(8.85 \times 10^{-12}\text{ C}^2/\text{N}\cdot\text{m}^2)(8.0 \times 10^{-2}\text{ m})} = -4.5 \times 10^5\text{ N/C},$$

or $|E(r = 8.0\text{ cm})| = 4.5 \times 10^5\text{ N/C}$.

(d) The minus sign indicates that $\vec{E}(r)$ points radially inward.

32. To evaluate the field using Gauss' law, we employ a cylindrical surface of area $2\pi r L$ where L is very large (large enough that contributions from the ends of the cylinder become irrelevant to the calculation). The volume within this surface is $V = \pi r^2 L$, or expressed more appropriate to our needs: $dV = 2\pi r L dr$. The charge enclosed is, with $A = 2.5 \times 10^{-6}\text{ C/m}^5$,

$$q_{\text{enc}} = \int_0^r A r^2 2\pi r L dr = \frac{\pi}{2} A L r^4.$$

By Gauss' law, we find $\Phi = |\vec{E}| (2\pi r L) = q_{\text{enc}} / \epsilon_0$; we thus obtain $|\vec{E}| = \frac{A r^3}{4 \epsilon_0}$.

(a) With $r = 0.030$ m, we find $|\vec{E}| = 1.9$ N/C.

(b) Once outside the cylinder, Eq. 23-12 is obeyed. To find $\lambda = q/L$ we must find the total charge q . Therefore,

$$\frac{q}{L} = \frac{1}{L} \int_0^{0.04} A r^2 2\pi r L dr = 1.0 \times 10^{-11} \text{ C/m}.$$

And the result, for $r = 0.050$ m, is $|\vec{E}| = \lambda / 2\pi\epsilon_0 r = 3.6$ N/C.

33. We use Eq. 23-13.

(a) To the left of the plates:

$$\vec{E} = (\sigma / 2\epsilon_0)(-\hat{i}) \text{ (from the right plate)} + (\sigma / 2\epsilon_0)\hat{i} \text{ (from the left one)} = 0.$$

(b) To the right of the plates:

$$\vec{E} = (\sigma / 2\epsilon_0)\hat{i} \text{ (from the right plate)} + (\sigma / 2\epsilon_0)(-\hat{i}) \text{ (from the left one)} = 0.$$

(c) Between the plates:

$$\begin{aligned} \vec{E} &= \left(\frac{\sigma}{2\epsilon_0} \right)(-\hat{i}) + \left(\frac{\sigma}{2\epsilon_0} \right)(-\hat{i}) = \left(\frac{\sigma}{\epsilon_0} \right)(-\hat{i}) = - \left(\frac{7.00 \times 10^{-22} \text{ C/m}^2}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2} \right) \hat{i} \\ &= (-7.91 \times 10^{-11} \text{ N/C}) \hat{i}. \end{aligned}$$

34. The charge distribution in this problem is equivalent to that of an infinite sheet of charge with surface charge density $\sigma = 4.50 \times 10^{-12} \text{ C/m}^2$ plus a small circular pad of radius $R = 1.80$ cm located at the middle of the sheet with charge density $-\sigma$. We denote the electric fields produced by the sheet and the pad with subscripts 1 and 2, respectively. Using Eq. 22-26 for $\vec{E}_2$, the net electric field $\vec{E}$ at a distance $z = 2.56$ cm along the central axis is then

$$\begin{aligned} \vec{E} &= \vec{E}_1 + \vec{E}_2 = \left(\frac{\sigma}{2\epsilon_0} \right) \hat{k} + \frac{(-\sigma)}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right) \hat{k} = \frac{\sigma z}{2\epsilon_0 \sqrt{z^2 + R^2}} \hat{k} \\ &= \frac{(4.50 \times 10^{-12} \text{ C/m}^2)(2.56 \times 10^{-2} \text{ m})}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2) \sqrt{(2.56 \times 10^{-2} \text{ m})^2 + (1.80 \times 10^{-2} \text{ m})^2}} \hat{k} = (0.208 \text{ N/C}) \hat{k}. \end{aligned}$$

35. In the region between sheets 1 and 2, the net field is $E_1 - E_2 + E_3 = 2.0 \times 10^5 \text{ N/C}$.

In the region between sheets 2 and 3, the net field is at its greatest value:

$$E_1 + E_2 + E_3 = 6.0 \times 10^5 \text{ N/C}.$$

The net field vanishes in the region to the right of sheet 3, where $E_1 + E_2 = E_3$. We note the implication that σ_3 is negative (and is the largest surface-density, in magnitude). These three conditions are sufficient for finding the fields:

$$E_1 = 1.0 \times 10^5 \text{ N/C}, \quad E_2 = 2.0 \times 10^5 \text{ N/C}, \quad E_3 = 3.0 \times 10^5 \text{ N/C}.$$

From Eq. 23-13, we infer (from these values of E)

$$\frac{|\sigma_3|}{|\sigma_2|} = \frac{3.0 \times 10^5 \text{ N/C}}{2.0 \times 10^5 \text{ N/C}} = 1.5.$$

Recalling our observation, above, about σ_3 , we conclude that $\frac{\sigma_3}{\sigma_2} = -1.5$.

36. According to Eq. 23-13 the electric field due to either sheet of charge with surface charge density $\sigma = 1.77 \times 10^{-22} \text{ C/m}^2$ is perpendicular to the plane of the sheet (pointing *away* from the sheet if the charge is positive) and has magnitude $E = \sigma/2\epsilon_0$. Using the superposition principle, we conclude:

(a) $E = \sigma/\epsilon_0 = (1.77 \times 10^{-22} \text{ C/m}^2)/(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2) = 2.00 \times 10^{-11} \text{ N/C}$, pointing in the upward direction, or $\vec{E} = (2.00 \times 10^{-11} \text{ N/C})\hat{j}$;

(b) $E = 0$;

(c) and, $E = \sigma/\epsilon_0$, pointing down, or $\vec{E} = -(2.00 \times 10^{-11} \text{ N/C})\hat{j}$.

37. **THINK** To calculate the electric field at a point very close to the center of a large, uniformly charged conducting plate, we replace the finite plate with an infinite plate having the same charge density. Planar symmetry then allows us to apply Gauss' law to calculate the electric field.

EXPRESS Using Gauss' law, we find the magnitude of the field to be $E = \sigma/\epsilon_0$, where σ is the area charge density for the surface just under the point. The charge is distributed uniformly over both sides of the original plate, with half being on the side near the field point. Thus, $\sigma = q/2A$.

ANALYZE (a) With $q = 6.0 \times 10^{-6} \text{ C}$ and $A = (0.080 \text{ m})^2$, we obtain

$$\sigma = \frac{q}{2A} = \frac{6.0 \times 10^{-6} \text{ C}}{2(0.080 \text{ m})^2} = 4.69 \times 10^{-4} \text{ C/m}^2.$$

The magnitude of the field is

$$E = \frac{\sigma}{\epsilon_0} = \frac{4.69 \times 10^{-4} \text{ C/m}^2}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2} = 5.3 \times 10^7 \text{ N/C}.$$

The field is normal to the plate and since the charge on the plate is positive, it points away from the plate.

(b) At a point far away from the plate, the electric field is nearly that of a point particle with charge equal to the total charge on the plate. The magnitude of the field is $E = q / 4\pi\epsilon_0 r^2 = kq / r^2$, where r is the distance from the plate. Thus,

$$E = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(6.0 \times 10^{-6} \text{ C})}{(30 \text{ m})^2} = 60 \text{ N/C}.$$

LEARN In summary, the electric field is nearly uniform ($E = \sigma / \epsilon_0$) close to the plate, but resembles that of a point charge far away from the plate.

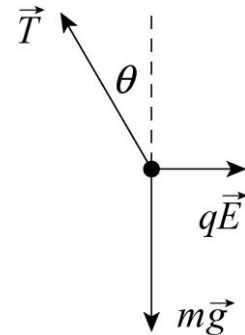
38. The field due to the sheet is $E = \frac{\sigma}{2\epsilon_0}$. The force (in magnitude) on the electron (due to that field) is $F = eE$, and assuming it's the *only* force then the acceleration is

$$a = \frac{e\sigma}{2\epsilon_0 m} = \text{slope of the graph} \quad (= 2.0 \times 10^5 \text{ m/s divided by } 7.0 \times 10^{-12} \text{ s}).$$

Thus we obtain $\sigma = 2.9 \times 10^{-6} \text{ C/m}^2$.

39. **THINK** Since the non-conducting charged ball is in equilibrium with the non-conducting charged sheet (see Fig. 23-49), both the vertical and horizontal components of the net force on the ball must be zero.

EXPRESS The forces acting on the ball are shown in the diagram to the right. The gravitational force has magnitude mg , where m is the mass of the ball; the electrical force has magnitude qE , where q is the charge on the ball and E is the magnitude of the electric field at the position of the ball; and the tension in the thread is denoted by T . The electric field produced by the plate is normal to the plate and points to the right. Since the ball is positively charged, the electric force on it also points to the right. The tension in the thread makes the angle θ ($= 30^\circ$) with the vertical. Since the ball is in



equilibrium the net force on it vanishes. The sum of the horizontal components yields

$$qE - T \sin \theta = 0$$

and the sum of the vertical components yields

$$T \cos \theta - mg = 0.$$

We solve for the electric field E and deduce σ , the charge density of the sheet, from $E = \sigma/2\epsilon_0$ (see Eq. 23-13).

ANALYZE The expression $T = qE/\sin \theta$, from the first equation, is substituted into the second to obtain $qE = mg \tan \theta$. The electric field produced by a large uniform sheet of charge is given by $E = \sigma/2\epsilon_0$, so

$$\frac{q\sigma}{2\epsilon_0} = mg \tan \theta$$

and we have

$$\begin{aligned}\sigma &= \frac{2\epsilon_0 mg \tan \theta}{q} = \frac{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(1.0 \times 10^{-6} \text{ kg})(9.8 \text{ m/s}^2) \tan 30^\circ}{2.0 \times 10^{-8} \text{ C}} \\ &= 5.0 \times 10^{-9} \text{ C/m}^2.\end{aligned}$$

LEARN Since both the sheet and the ball are positively charged, the force between them is repulsive. This is balanced by the horizontal component of the tension in the thread. The angle the thread makes with the vertical direction increases with the charge density of the sheet.

40. The point where the individual fields cancel cannot be in the region between the sheet and the particle ($-d < x < 0$) since the sheet and the particle have opposite-signed charges. The point(s) could be in the region to the right of the particle ($x > 0$) and in the region to the left of the sheet ($x < -d$); this is where the condition

$$\frac{|\sigma|}{2\epsilon_0} = \frac{Q}{4\pi\epsilon_0 r^2}$$

must hold. Solving this with the given values, we find $r = x = \pm\sqrt{3/2\pi} \approx \pm 0.691 \text{ m}$.

If $d = 0.20 \text{ m}$ (which is less than the magnitude of r found above), then neither of the points ($x \approx \pm 0.691 \text{ m}$) is in the “forbidden region” between the particle and the sheet. Thus, both values are allowed. Thus, we have

(a) $x = 0.691 \text{ m}$ on the positive axis, and

(b) $x = -0.691$ m on the negative axis.

(c) If, however, $d = 0.80$ m (greater than the magnitude of r found above), then one of the points ($x \approx -0.691$ m) is in the “forbidden region” between the particle and the sheet and is disallowed. In this part, the fields cancel only at the point $x \approx +0.691$ m.

41. The charge on the metal plate, which is negative, exerts a force of repulsion on the electron and stops it. First find an expression for the acceleration of the electron, then use kinematics to find the stopping distance. We take the initial direction of motion of the electron to be positive. Then, the electric field is given by $E = \sigma/\epsilon_0$, where σ is the surface charge density on the plate. The force on the electron is $F = -eE = -e\sigma/\epsilon_0$ and the acceleration is

$$a = \frac{F}{m} = -\frac{e\sigma}{\epsilon_0 m}$$

where m is the mass of the electron. The force is constant, so we use constant acceleration kinematics. If v_0 is the initial velocity of the electron, v is the final velocity, and x is the distance traveled between the initial and final positions, then $v^2 - v_0^2 = 2ax$. Set $v = 0$ and replace a with $-e\sigma/\epsilon_0 m$, then solve for x . We find

$$x = -\frac{v_0^2}{2a} = \frac{\epsilon_0 m v_0^2}{2e\sigma}.$$

Now $\frac{1}{2}mv_0^2$ is the initial kinetic energy K_0 , so

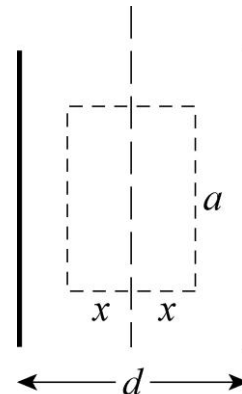
$$x = \frac{\epsilon_0 K_0}{e\sigma} = \frac{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(1.60 \times 10^{-17} \text{ J})}{(1.60 \times 10^{-19} \text{ C})(2.0 \times 10^{-6} \text{ C/m}^2)} = 4.4 \times 10^{-4} \text{ m}.$$

42. The surface charge density is given by

$$E = \sigma/\epsilon_0 \Rightarrow \sigma = \epsilon_0 E = (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(55 \text{ N/C}) = 4.9 \times 10^{-10} \text{ C/m}^2.$$

Since the area of the plates is $A = 1.0 \text{ m}^2$, the magnitude of the charge on the plate is $Q = \sigma A = 4.9 \times 10^{-10} \text{ C}$.

43. We use a Gaussian surface in the form of a box with rectangular sides. The cross section is shown with dashed lines in the diagram to the right. It is centered at the central plane of the slab, so the left and right faces are each a distance x from the central plane. We take the thickness of the rectangular solid to be a , the same as its length, so the left and right faces are squares.



The electric field is normal to the left and right faces and is uniform over them. Since $\rho = 5.80 \text{ fC/m}^3$ is positive, it points outward at both faces: toward the left at the left face and toward the right at the right face. Furthermore, the magnitude is the same at both faces. The electric flux through each of these faces is Ea^2 . The field is parallel to the other faces of the Gaussian surface and the flux through them is zero. The total flux through the Gaussian surface is $\Phi = 2Ea^2$. The volume enclosed by the Gaussian surface is $2a^2x$ and the charge contained within it is $q = 2a^2x\rho$. Gauss' law yields

$$2\varepsilon_0 Ea^2 = 2a^2x\rho.$$

We solve for the magnitude of the electric field: $E = \rho x / \varepsilon_0$.

(a) For $x = 0$, $E = 0$.

(b) For $x = 2.00 \text{ mm} = 2.00 \times 10^{-3} \text{ m}$,

$$E = \frac{\rho x}{\varepsilon_0} = \frac{(5.80 \times 10^{-15} \text{ C/m}^3)(2.00 \times 10^{-3} \text{ m})}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2} = 1.31 \times 10^{-6} \text{ N/C}.$$

(c) For $x = d/2 = 4.70 \text{ mm} = 4.70 \times 10^{-3} \text{ m}$,

$$E = \frac{\rho x}{\varepsilon_0} = \frac{(5.80 \times 10^{-15} \text{ C/m}^3)(4.70 \times 10^{-3} \text{ m})}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2} = 3.08 \times 10^{-6} \text{ N/C}.$$

(d) For $x = 26.0 \text{ mm} = 2.60 \times 10^{-2} \text{ m}$, we take a Gaussian surface of the same shape and orientation, but with $x > d/2$, so the left and right faces are outside the slab. The total flux through the surface is again $\Phi = 2Ea^2$ but the charge enclosed is now $q = a^2d\rho$. Gauss' law yields $2\varepsilon_0 Ea^2 = a^2d\rho$, so

$$E = \frac{\rho d}{2\varepsilon_0} = \frac{(5.80 \times 10^{-15} \text{ C/m}^3)(9.40 \times 10^{-3} \text{ m})}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 3.08 \times 10^{-6} \text{ N/C}.$$

44. We determine the (total) charge on the ball by examining the maximum value ($E = 5.0 \times 10^7 \text{ N/C}$) shown in the graph (which occurs at $r = 0.020 \text{ m}$). Thus, from $E = q / 4\pi\varepsilon_0 r^2$, we obtain

$$q = 4\pi\varepsilon_0 r^2 E = \frac{(0.020 \text{ m})^2 (5.0 \times 10^7 \text{ N/C})}{8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2} = 2.2 \times 10^{-6} \text{ C}.$$

45. (a) Since $r_1 = 10.0 \text{ cm} < r = 12.0 \text{ cm} < r_2 = 15.0 \text{ cm}$,

$$E(r) = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(4.00 \times 10^{-8} \text{ C})}{(0.120 \text{ m})^2} = 2.50 \times 10^4 \text{ N/C}.$$

(b) Since $r_1 < r_2 < r = 20.0 \text{ cm}$,

$$E(r) = \frac{1}{4\pi\epsilon_0} \frac{q_1 + q_2}{r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(4.00 + 2.00)(1 \times 10^{-8} \text{ C})}{(0.200 \text{ m})^2} = 1.35 \times 10^4 \text{ N/C}.$$

46. The field at the proton's location (but not caused by the proton) has magnitude E . The proton's charge is e . The ball's charge has magnitude q . Thus, as long as the proton is at $r \geq R$ then the force on the proton (caused by the ball) has magnitude

$$F = eE = e \left(\frac{q}{4\pi\epsilon_0 r^2} \right) = \frac{eq}{4\pi\epsilon_0 r^2}$$

where r is measured from the center of the ball (to the proton). This agrees with Coulomb's law from Chapter 22. We note that if $r = R$ then this expression becomes

$$F_R = \frac{eq}{4\pi\epsilon_0 R^2}.$$

(a) If we require $F = \frac{1}{2}F_R$, and solve for r , we obtain $r = \sqrt{2} R$. Since the problem asks for the measurement from the surface then the answer is $\sqrt{2} R - R = 0.41R$.

(b) Now we require $F_{\text{inside}} = \frac{1}{2}F_R$ where $F_{\text{inside}} = eE_{\text{inside}}$ and E_{inside} is given by Eq. 23-20. Thus,

$$e \left(\frac{q}{4\pi\epsilon_0 R^2} \right) r = \frac{1}{2} \frac{eq}{4\pi\epsilon_0 R^2} \quad \Rightarrow \quad r = \frac{1}{2} R = 0.50 R.$$

47. **THINK** The unknown charge is distributed uniformly over the surface of the conducting solid sphere.

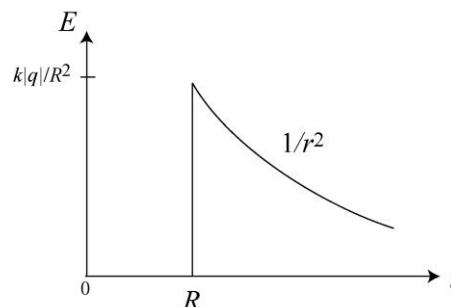
EXPRESS The electric field produced by the unknown charge at points outside the sphere is like the field of a point particle with charge equal to the net charge on the sphere. That is, the magnitude of the field is given by $E = |q|/4\pi\epsilon_0 r^2$, where $|q|$ is the magnitude of the charge on the sphere and r is the distance from the center of the sphere to the point where the field is measured.

ANALYZE Thus, we have

$$|q| = 4\pi\epsilon_0 r^2 E = \frac{(0.15 \text{ m})^2 (3.0 \times 10^3 \text{ N/C})}{8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2} = 7.5 \times 10^{-9} \text{ C}.$$

The field points inward, toward the sphere center, so the charge is negative, i.e., $q = -7.5 \times 10^{-9} \text{ C}$.

LEARN The electric field strength as a function of r is shown to the right. Inside the metal sphere, $E = 0$; outside the sphere, $E = k|q|/r^2$, where $k = 1/4\pi\epsilon_0$.



48. Let E_A designate the magnitude of the field at $r = 2.4 \text{ cm}$. Thus $E_A = 2.0 \times 10^7 \text{ N/C}$, and is totally due to the particle. Since $E_{\text{particle}} = q/4\pi\epsilon_0 r^2$, then the field due to the particle at any other point will relate to E_A by a ratio of distances squared. Now, we note that at $r = 3.0 \text{ cm}$ the total contribution (from particle and shell) is $8.0 \times 10^7 \text{ N/C}$. Therefore,

$$E_{\text{shell}} + E_{\text{particle}} = E_{\text{shell}} + (2.4/3)^2 E_A = 8.0 \times 10^7 \text{ N/C}.$$

Using the value for E_A noted above, we find $E_{\text{shell}} = 6.6 \times 10^7 \text{ N/C}$. Thus, with $r = 0.030 \text{ m}$, we find the charge Q using $E_{\text{shell}} = Q/4\pi\epsilon_0 r^2$:

$$Q = 4\pi\epsilon_0 r^2 E_{\text{shell}} = \frac{r^2 E_{\text{shell}}}{k} = \frac{(0.030 \text{ m})^2 (6.6 \times 10^7 \text{ N/C})}{8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2} = 6.6 \times 10^{-6} \text{ C}$$

49. At all points where there is an electric field, it is radially outward. For each part of the problem, use a Gaussian surface in the form of a sphere that is concentric with the sphere of charge and passes through the point where the electric field is to be found. The field is uniform on the surface, so $\oint \vec{E} \cdot d\vec{A} = 4\pi r^2 E$, where r is the radius of the Gaussian surface.

For $r < a$, the charge enclosed by the Gaussian surface is $q_1(r/a)^3$. Gauss' law yields

$$4\pi r^2 E = \left(\frac{q_1}{\epsilon_0} \right) \left(\frac{r}{a} \right)^3 \Rightarrow E = \frac{q_1 r}{4\pi\epsilon_0 a^3}.$$

(a) For $r = 0$, the above equation implies $E = 0$.

(b) For $r = a/2$, we have

$$E = \frac{q_1(a/2)}{4\pi\epsilon_0 a^3} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(5.00 \times 10^{-15} \text{ C})}{2(2.00 \times 10^{-2} \text{ m})^2} = 5.62 \times 10^{-2} \text{ N/C}.$$

(c) For $r = a$, we have

$$E = \frac{q_1}{4\pi\epsilon_0 a^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(5.00 \times 10^{-15} \text{ C})}{(2.00 \times 10^{-2} \text{ m})^2} = 0.112 \text{ N/C}.$$

In the case where $a < r < b$, the charge enclosed by the Gaussian surface is q_1 , so Gauss' law leads to

$$4\pi r^2 E = \frac{q_1}{\epsilon_0} \Rightarrow E = \frac{q_1}{4\pi\epsilon_0 r^2}.$$

(d) For $r = 1.50a$, we have

$$E = \frac{q_1}{4\pi\epsilon_0 r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(5.00 \times 10^{-15} \text{ C})}{(1.50 \times 2.00 \times 10^{-2} \text{ m})^2} = 0.0499 \text{ N/C}.$$

(e) In the region $b < r < c$, since the shell is conducting, the electric field is zero. Thus, for $r = 2.30a$, we have $E = 0$.

(f) For $r > c$, the charge enclosed by the Gaussian surface is zero. Gauss' law yields $4\pi r^2 E = 0 \Rightarrow E = 0$. Thus, $E = 0$ at $r = 3.50a$.

(g) Consider a Gaussian surface that lies completely within the conducting shell. Since the electric field is everywhere zero on the surface, $\oint \vec{E} \cdot d\vec{A} = 0$ and, according to Gauss' law, the net charge enclosed by the surface is zero. If Q_i is the charge on the inner surface of the shell, then $q_1 + Q_i = 0$ and $Q_i = -q_1 = -5.00 \text{ fC}$.

(h) Let Q_o be the charge on the outer surface of the shell. Since the net charge on the shell is $-q$, $Q_i + Q_o = -q_1$. This means

$$Q_o = -q_1 - Q_i = -q_1 - (-q_1) = 0.$$

50. The point where the individual fields cancel cannot be in the region between the shells since the shells have opposite-signed charges. It cannot be inside the radius R of one of the shells since there is only one field contribution there (which would not be canceled by another field contribution and thus would not lead to zero net field). We note shell 2 has greater magnitude of charge ($|\sigma_2|A_2$) than shell 1, which implies the point is not to the right of shell 2 (any such point would always be closer to the larger charge and thus no possibility for cancellation of equal-magnitude fields could occur). Consequently, the point should be in the region to the left of shell 1 (at a distance $r > R_1$ from its center); this is where the condition

$$E_1 = E_2 \Rightarrow \frac{|q_1|}{4\pi\epsilon_0 r^2} = \frac{|q_2|}{4\pi\epsilon_0 (r+L)^2}$$

or

$$\frac{\sigma_1 A_1}{4\pi\epsilon_0 r^2} = \frac{|\sigma_2| A_2}{4\pi\epsilon_0 (r+L)^2}.$$

Using the fact that the area of a sphere is $A = 4\pi R^2$, this condition simplifies to

$$r = \frac{L}{(R_2/R_1)\sqrt{|\sigma_2|/\sigma_1} - 1} = 3.3 \text{ cm}.$$

We note that this value satisfies the requirement $r > R_1$. The answer, then, is that the net field vanishes at $x = -r = -3.3 \text{ cm}$.

51. **THINK** Since our system possesses spherical symmetry, to calculate the electric field strength, we may apply Gauss' law and take the Gaussian surface to be in the form of a sphere of radius r .

EXPRESS To find an expression for the electric field inside the shell in terms of A and the distance from the center of the shell, choose A so the field does not depend on the distance. We use a Gaussian surface in the form of a sphere with radius r_g , concentric with the spherical shell and within it ($a < r_g < b$). Gauss' law will be used to find the magnitude of the electric field a distance r_g from the shell center. The charge that is both in the shell and within the Gaussian sphere is given by the integral $q_s = \int \rho dV$ over the portion of the shell within the Gaussian surface. Since the charge distribution has spherical symmetry, we may take dV to be the volume of a spherical shell with radius r and infinitesimal thickness dr : $dV = 4\pi r^2 dr$. Thus,

$$q_s = 4\pi \int_a^{r_g} \rho r^2 dr = 4\pi \int_a^{r_g} \frac{A}{r} r^2 dr = 4\pi A \int_a^{r_g} r dr = 2\pi A (r_g^2 - a^2).$$

The total charge inside the Gaussian surface is

$$q_{\text{enc}} = q + q_s = q + 2\pi A(r_g^2 - a^2).$$

The electric field is radial, so the flux through the Gaussian surface is $\Phi = 4\pi r_g^2 E$, where E is the magnitude of the field. Gauss' law yields

$$\Phi = q_{\text{enc}} / \epsilon_0 \Rightarrow 4\pi \epsilon_0 E r_g^2 = q + 2\pi A(r_g^2 - a^2).$$

We solve for E :

$$E = \frac{1}{4\pi \epsilon_0} \left[\frac{q}{r_g^2} + 2\pi A - \frac{2\pi A a^2}{r_g^2} \right]$$

ANALYZE For the field to be uniform, the first and last terms in the brackets must cancel. They do if $q - 2\pi A a^2 = 0$ or $A = q/2\pi a^2$. With $a = 2.00 \times 10^{-2} \text{ m}$ and $q = 45.0 \times 10^{-15} \text{ C}$, we have $A = 1.79 \times 10^{-11} \text{ C/m}^2$.

LEARN The value we have found for A ensures the uniformity of the field strength inside the shell. Using the result found above, we can readily show that the electric field in the region $a \leq r \leq b$ is

$$E = \frac{2\pi A}{4\pi\epsilon_0} = \frac{A}{2\epsilon_0} = \frac{1.79 \times 10^{-11} \text{ C/m}^2}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 1.01 \text{ N/C}.$$

52. The field is zero for $0 \leq r \leq a$ as a result of Eq. 23-16. Thus,

(a) $E = 0$ at $r = 0$,

(b) $E = 0$ at $r = a/2.00$, and

(c) $E = 0$ at $r = a$.

For $a \leq r \leq b$ the enclosed charge q_{enc} (for $a \leq r \leq b$) is related to the volume by

$$q_{\text{enc}} = \rho \left[\frac{4\pi r^3}{3} - \frac{4\pi a^3}{3} \right].$$

Therefore, the electric field is

$$E = \frac{1}{4\pi\epsilon_0} \frac{q_{\text{enc}}}{r^2} = \frac{\rho}{4\pi\epsilon_0 r^2} \left[\frac{4\pi r^3}{3} - \frac{4\pi a^3}{3} \right] = \frac{\rho}{3\epsilon_0} \frac{r^3 - a^3}{r^2}$$

for $a \leq r \leq b$.

(d) For $r = 1.50a$, we have

$$E = \frac{\rho}{3\epsilon_0} \frac{(1.50a)^3 - a^3}{(1.50a)^2} = \frac{\rho a}{3\epsilon_0} \left(\frac{2.375}{2.25} \right) = \frac{(1.84 \times 10^{-9} \text{ C/m}^3)(0.100 \text{ m})}{3(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} \left(\frac{2.375}{2.25} \right) = 7.32 \text{ N/C}.$$

(e) For $r = b = 2.00a$, the electric field is

$$E = \frac{\rho}{3\epsilon_0} \frac{(2.00a)^3 - a^3}{(2.00a)^2} = \frac{\rho a}{3\epsilon_0} \left(\frac{7}{4} \right) = \frac{(1.84 \times 10^{-9} \text{ C/m}^3)(0.100 \text{ m})}{3(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} \left(\frac{7}{4} \right) = 12.1 \text{ N/C}.$$

(f) For $r \geq b$ we have $E = q_{\text{total}} / 4\pi\epsilon_0 r^2$ or

$$E = \frac{\rho}{3\epsilon_0} \frac{b^3 - a^3}{r^2}.$$

Thus, for $r = 3.00b = 6.00a$, the electric field is

$$E = \frac{\rho}{3\epsilon_0} \frac{(2.00a)^3 - a^3}{(6.00a)^2} = \frac{\rho a}{3\epsilon_0} \left(\frac{7}{36} \right) = \frac{(1.84 \times 10^{-9} \text{ C/m}^3)(0.100 \text{ m})}{3(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} \left(\frac{7}{36} \right) = 1.35 \text{ N/C}.$$

53. (a) We integrate the volume charge density over the volume and require the result be equal to the total charge:

$$\int dx \int dy \int dz \rho = 4\pi \int_0^R dr r^2 \rho = Q.$$

Substituting the expression $\rho = \rho_s r/R$, with $\rho_s = 14.1 \text{ pC/m}^3$, and performing the integration leads to

$$4\pi \left(\frac{\rho_s}{R} \right) \left(\frac{R^4}{4} \right) = Q$$

or

$$Q = \pi \rho_s R^3 = \pi (14.1 \times 10^{-12} \text{ C/m}^3) (0.0560 \text{ m})^3 = 7.78 \times 10^{-15} \text{ C}.$$

(b) At $r = 0$, the electric field is zero ($E = 0$) since the enclosed charge is zero.

At a certain point within the sphere, at some distance r from the center, the field (see Eq. 23-8 through Eq. 23-10) is given by Gauss' law:

$$E = \frac{1}{4\pi\epsilon_0} \frac{q_{\text{enc}}}{r^2}$$

where q_{enc} is given by an integral similar to that worked in part (a):

$$q_{\text{enc}} = 4\pi \int_0^r dr r^2 \rho = 4\pi \left(\frac{\rho_s}{R} \right) \left(\frac{r^4}{4} \right).$$

Therefore,

$$E = \frac{1}{4\pi\epsilon_0} \frac{\pi \rho_s r^4}{R r^2} = \frac{1}{4\pi\epsilon_0} \frac{\pi \rho_s r^2}{R}.$$

(c) For $r = R/2.00$, where $R = 5.60 \text{ cm}$, the electric field is

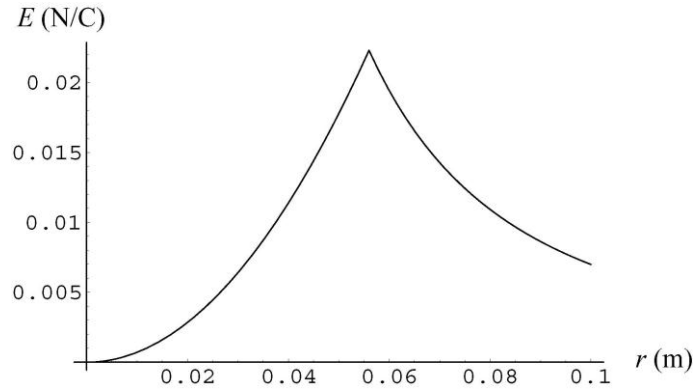
$$\begin{aligned} E &= \frac{1}{4\pi\epsilon_0} \frac{\pi \rho_s (R/2.00)^2}{R} = \frac{1}{4\pi\epsilon_0} \frac{\pi \rho_s R}{4.00} \\ &= \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \pi (14.1 \times 10^{-12} \text{ C/m}^3) (0.0560 \text{ m})}{4.00} \\ &= 5.58 \times 10^{-3} \text{ N/C}. \end{aligned}$$

(d) For $r = R$, the electric field is

$$E = \frac{1}{4\pi\epsilon_0} \frac{\pi\rho_s R^2}{R} = \frac{\pi\rho_s R}{4\pi\epsilon_0} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \pi (14.1 \times 10^{-12} \text{ C/m}^3) (0.0560 \text{ m})$$

$$= 2.23 \times 10^{-2} \text{ N/C}.$$

(e) The electric field strength as a function of r is depicted below:



54. Applying Eq. 23-20, we have

$$E_1 = \frac{|q_1|}{4\pi\epsilon_0 R^3} r_1 = \frac{|q_1|}{4\pi\epsilon_0 R^3} \left(\frac{R}{2} \right) = \frac{1}{2} \frac{|q_1|}{4\pi\epsilon_0 R^2}.$$

Also, outside sphere 2 we have

$$E_2 = \frac{|q_2|}{4\pi\epsilon_0 r^2} = \frac{|q_2|}{4\pi\epsilon_0 (1.50R)^2}.$$

Equating these and solving for the ratio of charges, we arrive at $\frac{q_2}{q_1} = \frac{9}{8} = 1.125$.

55. We use

$$E(r) = \frac{q_{\text{enc}}}{4\pi\epsilon_0 r^2} = \frac{1}{4\pi\epsilon_0 r^2} \int_0^r \rho(r) 4\pi r^2 dr$$

to solve for $\rho(r)$ and obtain

$$\rho(r) = \frac{\epsilon_0}{r^2} \frac{d}{dr} r^2 E(r) = \frac{\epsilon_0}{r^2} \frac{d}{dr} (Kr^6) = 6K\epsilon_0 r^3.$$

56. (a) There is no flux through the sides, so we have two contributions to the flux, one from the $x = 2$ end (with $\Phi_2 = +(2 + 2)(\pi(0.20)^2) = 0.50 \text{ N} \cdot \text{m}^2/\text{C}$) and one from the $x = 0$ end (with $\Phi_0 = -(2)(\pi(0.20)^2)$).

(b) By Gauss' law we have $q_{\text{enc}} = \epsilon_0 (\Phi_2 + \Phi_0) = 2.2 \times 10^{-12} \text{ C}$.

57. (a) For $r < R$, $E = 0$ (see Eq. 23-16).

(b) For r slightly greater than R ,

$$E_R = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \approx \frac{q}{4\pi\epsilon_0 R^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(2.00 \times 10^{-7} \text{ C})}{(0.250 \text{ m})^2} = 2.88 \times 10^4 \text{ N/C}.$$

(c) For $r > R$, $E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = E_R \left(\frac{R}{r}\right)^2 = (2.88 \times 10^4 \text{ N/C}) \left(\frac{0.250 \text{ m}}{3.00 \text{ m}}\right)^2 = 200 \text{ N/C}.$

58. From Gauss's law, we have

$$\Phi = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{\sigma \pi r^2}{\epsilon_0} = \frac{(8.0 \times 10^{-9} \text{ C/m}^2) \pi (0.050 \text{ m})^2}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2} = 7.1 \text{ N} \cdot \text{m}^2/\text{C}.$$

59. (a) At $x = 0.040 \text{ m}$, the net field has a rightward ($+x$) contribution (computed using Eq. 23-13) from the charge lying between $x = -0.050 \text{ m}$ and $x = 0.040 \text{ m}$, and a leftward ($-x$) contribution (again computed using Eq. 23-13) from the charge in the region from $x = 0.040 \text{ m}$ to $x = 0.050 \text{ m}$. Thus, since $\sigma = q/A = \rho V/A = \rho \Delta x$ in this situation, we have

$$|\vec{E}| = \frac{\rho(0.090 \text{ m})}{2\epsilon_0} - \frac{\rho(0.010 \text{ m})}{2\epsilon_0} = \frac{(1.2 \times 10^{-9} \text{ C/m}^3)(0.090 \text{ m} - 0.010 \text{ m})}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 5.4 \text{ N/C}.$$

(b) In this case, the field contributions from all layers of charge point rightward, and we obtain

$$|\vec{E}| = \frac{\rho(0.100 \text{ m})}{2\epsilon_0} = \frac{(1.2 \times 10^{-9} \text{ C/m}^3)(0.100 \text{ m})}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 6.8 \text{ N/C}.$$

60. (a) We consider the radial field produced at points within a uniform cylindrical distribution of charge. The volume enclosed by a Gaussian surface in this case is $L\pi r^2$. Thus, Gauss' law leads to

$$E = \frac{|q_{\text{enc}}|}{\epsilon_0 A_{\text{cylinder}}} = \frac{|\rho|(L\pi r^2)}{\epsilon_0 (2\pi r L)} = \frac{|\rho|r}{2\epsilon_0}.$$

(b) We note from the above expression that the magnitude of the radial field grows with r .

(c) Since the charged powder is negative, the field points radially inward.

(d) The largest value of r that encloses charged material is $r_{\text{max}} = R$. Therefore, with $|\rho| = 0.0011 \text{ C/m}^3$ and $R = 0.050 \text{ m}$, we obtain

$$E_{\max} = \frac{|\rho|R}{2\epsilon_0} = \frac{(0.0011 \text{ C/m}^3)(0.050 \text{ m})}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2)} = 3.1 \times 10^6 \text{ N/C}.$$

(e) According to condition 1 mentioned in the problem, the field is high enough to produce an electrical discharge (at $r = R$).

61. **THINK** Our system consists of two concentric metal shells. We apply the superposition principle and Gauss' law to calculate the electric field everywhere.

EXPRESS At all points where there is an electric field, it is radially outward. For each part of the problem, use a Gaussian surface in the form of a sphere that is concentric with the metal shells of charge and passes through the point where the electric field is to be found. The field is uniform on the surface, so

$$\Phi = \oint \vec{E} \cdot d\vec{A} = 4\pi r^2 E = \frac{q_{\text{enc}}}{\epsilon_0},$$

where r is the radius of the Gaussian surface.

ANALYZE (a) For $r < a$, the charge enclosed is $q_{\text{enc}} = 0$, so $E = 0$ in the region inside the shell.

(b) For $a < r < b$, the charged enclosed by the Gaussian surface is $q_{\text{enc}} = q_a$, so the field strength is $E = q_a / 4\pi\epsilon_0 r^2$.

(c) For $r > b$, the charged enclosed by the Gaussian surface is $q_{\text{enc}} = q_a + q_b$, so the field strength is $E = (q_a + q_b) / 4\pi\epsilon_0 r^2$.

(d) Since $E = 0$ for $r < a$ the charge on the inner surface of the inner shell is always zero. The charge on the outer surface of the inner shell is therefore q_a . Since $E = 0$ inside the metallic outer shell the net charge enclosed in a Gaussian surface that lies in between the inner and outer surfaces of the outer shell is zero. Thus the inner surface of the outer shell must carry a charge $-q_a$, leaving the charge on the outer surface of the outer shell to be $q_b + q_a$.

LEARN The concepts involved in this problem are discussed in Section 23-9 of the text. In the case of a single shell of radius R and charge q , the field strength is $E = 0$ for $r < R$, and $E = q / 4\pi\epsilon_0 r^2$ for $r > R$ (see Eqs. 23-15 and 23-16).

62. (a) The direction of the electric field at P_1 is away from q_1 and its magnitude is

$$|\vec{E}| = \frac{q}{4\pi\epsilon_0 r_1^2} = \frac{(8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)(1.0 \times 10^{-7} \text{ C})}{(0.015 \text{ m})^2} = 4.0 \times 10^6 \text{ N/C}.$$

(b) $\vec{E} = 0$, since P_2 is inside the metal.

63. The proton is in uniform circular motion, with the electrical force of the sphere on the proton providing the centripetal force. According to Newton's second law, $F = mv^2/r$, where F is the magnitude of the force, v is the speed of the proton, and r is the radius of its orbit, essentially the same as the radius of the sphere. The magnitude of the force on the proton is $F = e|q|/4\pi\epsilon_0 r^2$, where $|q|$ is the magnitude of the charge on the sphere. Thus,

$$\frac{1}{4\pi\epsilon_0} \frac{e|q|}{r^2} = \frac{mv^2}{r}$$

so

$$|q| = \frac{4\pi\epsilon_0 mv^2 r}{e} = \frac{(1.67 \times 10^{-27} \text{ kg})(3.00 \times 10^5 \text{ m/s})^2 (0.0100 \text{ m})}{(8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)(1.60 \times 10^{-9} \text{ C})} = 1.04 \times 10^{-9} \text{ C}.$$

The force must be inward, toward the center of the sphere, and since the proton is positively charged, the electric field must also be inward. The charge on the sphere is negative: $q = -1.04 \times 10^{-9} \text{ C}$.

64. We interpret the question as referring to the field *just* outside the sphere (that is, at locations roughly equal to the radius r of the sphere). Since the area of a sphere is $A = 4\pi r^2$ and the surface charge density is $\sigma = q/A$ (where we assume q is positive for brevity), then

$$E = \frac{\sigma}{\epsilon_0} = \frac{1}{\epsilon_0} \left(\frac{q}{4\pi r^2} \right) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

which we recognize as the field of a point charge (see Eq. 22-3).

65. (a) Since the volume contained within a radius of $\frac{1}{2}R$ is one-eighth the volume contained within a radius of R , the charge at $0 < r < R/2$ is $Q/8$. The fraction is $1/8 = 0.125$.

(b) At $r = R/2$, the magnitude of the field is

$$E = \frac{Q/8}{4\pi\epsilon_0 (R/2)^2} = \frac{1}{2} \frac{Q}{4\pi\epsilon_0 R^2}$$

and is equivalent to *half* the field at the surface. Thus, the ratio is 0.500.

66. (a) The flux is still $-750 \text{ N}\cdot\text{m}^2/\text{C}$, since it depends only on the amount of charge enclosed.

(b) We use $\Phi = q / \epsilon_0$ to obtain the charge q :

$$q = \epsilon_0 \Phi = (8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2) (-750 \text{ N}\cdot\text{m}^2/\text{C}) = -6.64 \times 10^{-9} \text{ C}.$$

67. **THINK** The electric field at P is due to the charge on the surface of the metallic conductor and the point charge Q .

EXPRESS The initial field (evaluated “just outside the outer surface” which means it is evaluated at $R_2 = 0.20 \text{ m}$, the outer radius of the conductor) is related to the charge q on the hollow conductor by Eq. 23-15: $E_{\text{initial}} = q / 4\pi\epsilon_0 R_2^2$. After the point charge Q is placed at the geometric center of the hollow conductor, the final field at that point is a combination of the initial and that due to Q (determined by Eq. 22-3):

$$E_{\text{final}} = E_{\text{initial}} + \frac{Q}{4\pi\epsilon_0 R_2^2}.$$

ANALYZE (a) The charge on the spherical shell is

$$q = 4\pi\epsilon_0 R_2^2 E_{\text{initial}} = \frac{(0.20 \text{ m})^2 (450 \text{ N/C})}{8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2} = 2.0 \times 10^{-9} \text{ C}.$$

(b) Similarly, using the equation above, we find the point charge to be

$$Q = 4\pi\epsilon_0 R_2^2 (E_{\text{final}} - E_{\text{initial}}) = \frac{(0.20 \text{ m})^2 (180 \text{ N/C} - 450 \text{ N/C})}{8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2} = -1.2 \times 10^{-9} \text{ C}.$$

(c) In order to cancel the field (due to Q) within the conducting material, there must be an amount of charge equal to $-Q$ distributed uniformly on the inner surface (of radius R_1). Thus, the answer is $+1.2 \times 10^{-9} \text{ C}$.

(d) Since the total excess charge on the conductor is q and is located on the surfaces, then the outer surface charge must equal the total minus the inner surface charge. Thus, the answer is $2.0 \times 10^{-9} \text{ C} - 1.2 \times 10^{-9} \text{ C} = +0.80 \times 10^{-9} \text{ C}$.

LEARN The key idea here is to realize that the electric field inside the conducting shell ($R_1 < r < R_2$) must be zero, so the charge must be distributed in such a way that the charge enclosed by a Gaussian sphere of radius r ($R_1 < r < R_2$) is zero.

68. Let $\Phi_0 = 10^3 \text{ N}\cdot\text{m}^2/\text{C}$. The net flux through the entire surface of the dice is given by

$$\Phi = \sum_{n=1}^6 \Phi_n = \sum_{n=1}^6 \left[-10^n n \Phi_0 \right] = \Phi_0 \left[-1 + 2 - 3 + 4 - 5 + 6 \right] = 3\Phi_0.$$

Thus, the net charge enclosed is

$$q = \epsilon_0 \Phi = 3\epsilon_0 \Phi_0 = 3(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(10^3 \text{ N} \cdot \text{m}^2/\text{C}) = 2.66 \times 10^{-8} \text{ C}.$$

69. Since the fields involved are uniform, the precise location of P is not relevant; what is important is it is above the three sheets, with the positively charged sheets contributing upward fields and the negatively charged sheet contributing a downward field, which conveniently conforms to usual conventions (of upward as positive and downward as negative). The net field is directed upward $(+\hat{j})$, and (from Eq. 23-13) its magnitude is

$$|\vec{E}| = \frac{\sigma_1}{2\epsilon_0} + \frac{\sigma_2}{2\epsilon_0} + \frac{\sigma_3}{2\epsilon_0} = \frac{1.0 \times 10^{-6} \text{ C/m}^2}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 5.65 \times 10^4 \text{ N/C}.$$

In unit-vector notation, we have $\vec{E} = (5.65 \times 10^4 \text{ N/C})\hat{j}$.

70. Since the charge distribution is uniform, we can find the total charge q by multiplying ρ by the spherical volume $(\frac{4}{3}\pi r^3)$ with $r = R = 0.050 \text{ m}$. This gives $q = 1.68 \text{ nC}$.

(a) Applying Eq. 23-20 with $r = 0.035 \text{ m}$, we have $E_{\text{internal}} = \frac{|q|r}{4\pi\epsilon_0 R^3} = 4.2 \times 10^3 \text{ N/C}$.

(b) Outside the sphere we have (with $r = 0.080 \text{ m}$)

$$E_{\text{external}} = \frac{|q|}{4\pi\epsilon_0 r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.68 \times 10^{-9} \text{ C})}{(0.080 \text{ m})^2} = 2.4 \times 10^3 \text{ N/C}.$$

71. We choose a coordinate system whose origin is at the center of the flat base, such that the base is in the xy plane and the rest of the hemisphere is in the $z > 0$ half space.

(a) $\Phi = \pi R^2 (-\hat{k}) \cdot E\hat{k} = -\pi R^2 E = -\pi(0.0568 \text{ m})^2(2.50 \text{ N/C}) = -0.0253 \text{ N} \cdot \text{m}^2/\text{C}$.

(b) Since the flux through the entire hemisphere is zero, the flux through the curved surface is $\vec{\Phi}_c = -\Phi_{\text{base}} = \pi R^2 E = 0.0253 \text{ N} \cdot \text{m}^2/\text{C}$.

72. The net enclosed charge q is given by

$$q = \epsilon_0 \Phi = (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(-48 \text{ N} \cdot \text{m}^2/\text{C}) = -4.2 \times 10^{-10} \text{ C}.$$

73. (a) From Gauss' law, we get $\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q_{\text{enc}}}{r^3} \vec{r} = \frac{1}{4\pi\epsilon_0} \frac{(4\pi\rho r^3/3)\vec{r}}{r^3} = \frac{\rho\vec{r}}{3\epsilon_0}$.

(b) The charge distribution in this case is equivalent to that of a whole sphere of charge density ρ plus a smaller sphere of charge density $-\rho$ that fills the void. By superposition

$$\vec{E}(\vec{r}) = \frac{\rho\vec{r}}{3\epsilon_0} + \frac{(-\rho)(\vec{r} - \vec{a})}{3\epsilon_0} = \frac{\rho\vec{a}}{3\epsilon_0}.$$

74. (a) The cube is totally within the spherical volume, so the charge enclosed is

$$q_{\text{enc}} = \rho V_{\text{cube}} = (500 \times 10^{-9} \text{ C/m}^3)(0.0400 \text{ m})^3 = 3.20 \times 10^{-11} \text{ C}.$$

By Gauss' law, we find $\Phi = q_{\text{enc}}/\epsilon_0 = 3.62 \text{ N}\cdot\text{m}^2/\text{C}$.

(b) Now the sphere is totally contained within the cube (note that the radius of the sphere is less than half the side-length of the cube). Thus, the total charge is

$$q_{\text{enc}} = \rho V_{\text{sphere}} = 4.5 \times 10^{-10} \text{ C}.$$

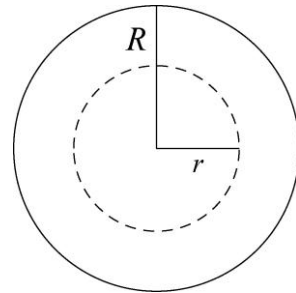
By Gauss' law, we find $\Phi = q_{\text{enc}}/\epsilon_0 = 51.1 \text{ N}\cdot\text{m}^2/\text{C}$.

75. The electric field is radially outward from the central wire. We want to find its magnitude in the region between the wire and the cylinder as a function of the distance r from the wire. Since the magnitude of the field at the cylinder wall is known, we take the Gaussian surface to coincide with the wall. Thus, the Gaussian surface is a cylinder with radius R and length L , coaxial with the wire. Only the charge on the wire is actually enclosed by the Gaussian surface; we denote it by q . The area of the Gaussian surface is $2\pi RL$, and the flux through it is $\Phi = 2\pi RLE$. We assume there is no flux through the ends of the cylinder, so this Φ is the total flux. Gauss' law yields $q = 2\pi\epsilon_0 RLE$. Thus,

$$q = 2\pi(8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2)(0.014 \text{ m})(0.16 \text{ m})(2.9 \times 10^4 \text{ N/C}) = 3.6 \times 10^{-9} \text{ C}.$$

76. (a) The diagram shows a cross section (or, perhaps more appropriately, "end view") of the charged cylinder (solid circle).

Consider a Gaussian surface in the form of a cylinder with radius r and length ℓ , coaxial with the charged cylinder. An "end view" of the Gaussian surface is shown as a dashed circle. The charge enclosed by it is $q = \rho V = \pi r^2 \ell \rho$, where $V = \pi r^2 \ell$ is the volume of the cylinder. If ρ is positive, the electric field lines are radially



outward, normal to the Gaussian surface and distributed uniformly along it. Thus, the total flux through the Gaussian cylinder is $\Phi = EA_{\text{cylinder}} = E(2\pi r\ell)$. Now, Gauss' law leads to

$$2\pi\epsilon_0 r\ell E = \pi r^2 \ell \rho \Rightarrow E = \frac{\rho r}{2\epsilon_0}.$$

(b) Next, we consider a cylindrical Gaussian surface of radius $r > R$. If the external field E_{ext} then the flux is $\Phi = 2\pi r\ell E_{\text{ext}}$. The charge enclosed is the total charge in a section of the charged cylinder with length ℓ . That is, $q = \pi R^2 \ell \rho$. In this case, Gauss' law yields

$$2\pi\epsilon_0 r\ell E_{\text{ext}} = \pi R^2 \ell \rho \Rightarrow E_{\text{ext}} = \frac{R^2 \rho}{2\epsilon_0 r}.$$

77. THINK The total charge on the conducting shell is equal to the sum of the charges on the shell's inner surface and the outer surface.

EXPRESS Let q_{in} be the charge on the inner surface and q_{out} the charge on the outer surface. The net charge on the shell is $Q = q_{\text{in}} + q_{\text{out}}$.

ANALYZE (a) In order to have net charge $Q = -10 \mu\text{C}$ when the charge on the outer surface is $q_{\text{out}} = -14 \mu\text{C}$, then there must be

$$q_{\text{in}} = Q - q_{\text{out}} = -10 \mu\text{C} - (-14 \mu\text{C}) = +4 \mu\text{C}$$

on the inner surface (since charges reside on the surfaces of a conductor in electrostatic situations).

(b) Let q be the charge of the particle. In order to cancel the electric field inside the conducting material, the contribution from the $q_{\text{in}} = +4 \mu\text{C}$ on the inner surface must be canceled by that of the charged particle in the hollow, that is, $q_{\text{enc}} = q + q_{\text{in}} = 0$. Thus, the particle's charge is $q = -q_{\text{in}} = -4 \mu\text{C}$.

LEARN The key idea here is to realize that the electric field inside the conducting shell must be zero. Thus, in the presence of a point charge in the hollow, the charge on the shell must be redistributed between its inner and outer surfaces in such a way that the net charge enclosed by a Gaussian sphere of radius r ($R_1 < r < R_2$, where R_1 is the inner radius and R_2 is the outer radius) remains zero.

78. (a) Outside the sphere, we use Eq. 23-15 and obtain

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(6.00 \times 10^{-12} \text{ C})}{(0.0600 \text{ m})^2} = 15.0 \text{ N/C}.$$

(b) With $q = +6.00 \times 10^{-12} \text{ C}$, Eq. 23-20 leads to $E = 25.3 \text{ N/C}$.

79. (a) The mass flux is $wd\rho v = (3.22 \text{ m})(1.04 \text{ m})(1000 \text{ kg/m}^3)(0.207 \text{ m/s}) = 693 \text{ kg/s}$.

(b) Since water flows only through area wd , the flux through the larger area is still 693 kg/s .

(c) Now the mass flux is $(wd/2)\rho v = (693 \text{ kg/s})/2 = 347 \text{ kg/s}$.

(d) Since the water flows through an area $(wd/2)$, the flux is 347 kg/s .

(e) Now the flux is $(wd \cos \theta)\rho v = (693 \text{ kg/s})(\cos 34^\circ) = 575 \text{ kg/s}$.

80. The field due to a sheet of charge is given by Eq. 23-13. Both sheets are horizontal (parallel to the xy plane), producing vertical fields (parallel to the z axis). At points above the $z = 0$ sheet (sheet A), its field points upward (toward $+z$); at points above the $z = 2.0$ sheet (sheet B), its field does likewise. However, below the $z = 2.0$ sheet, its field is oriented downward.

(a) The magnitude of the net field in the region between the sheets is

$$|\vec{E}| = \frac{\sigma_A}{2\epsilon_0} - \frac{\sigma_B}{2\epsilon_0} = \frac{8.00 \times 10^{-9} \text{ C/m}^2 - 3.00 \times 10^{-9} \text{ C/m}^2}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 2.82 \times 10^2 \text{ N/C}.$$

(b) The magnitude of the net field at points above both sheets is

$$|\vec{E}| = \frac{\sigma_A}{2\epsilon_0} + \frac{\sigma_B}{2\epsilon_0} = \frac{8.00 \times 10^{-9} \text{ C/m}^2 + 3.00 \times 10^{-9} \text{ C/m}^2}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 6.21 \times 10^2 \text{ N/C}.$$

81. (a) The field maximum occurs at the outer surface:

$$E_{\text{max}} = \left(\frac{|q|}{4\pi\epsilon_0 r^2} \right)_{\text{at } r=R} = \frac{|q|}{4\pi\epsilon_0 R^2}$$

Applying Eq. 23-20, we have

$$E_{\text{internal}} = \frac{|q|}{4\pi\epsilon_0 R^3} r = \frac{1}{4} E_{\text{max}} \Rightarrow r = \frac{R}{4} = 0.25 R.$$

(b) Outside sphere 2 we have

$$E_{\text{external}} = \frac{|q|}{4\pi\epsilon_0 r^2} = \frac{1}{4} E_{\text{max}} \Rightarrow r = 2.0R.$$